AAROGYA VAANI

MVP Implementation Plan

Focused Use Case · Tech Stack · 3-Month Delivery | Jun – Aug 2026

1	4	13	2+
Core Use Case	Tech Components	Weeks to Deliver	Pilot CHO Sites

THE PROBLEM WE ARE SOLVING

Context for the MVP scope

1:10,000

Doctor-patient ratio
in rural India

40–80

Patients a CHO sees
per day with no tools

0%

Patient notes
are digitised today

No Decision Support

CHOs diagnose under crushing cognitive load — no clinical prompts, no guidelines at hand.

No Digital Records

Hand-written registers mean zero continuity of care when a patient moves between facilities.

Wrong Escalations

Hospitals clogged with manageable cases while critical referrals get missed entirely.

THE MVP USE CASE — VOICE-GUIDED TRIAGE FOR FEVER & RESPIRATORY ILLNESS

Chosen for high prevalence, clear protocol, and minimum integration complexity

WHY THIS USE CASE?

1. **70% of CHO caseload**
   Fever, cough, breathlessness are the most common presenting complaints — maximum real-world impact.
2. **Clear WHO protocol**
   IMCI triage for ARI and fever is a well-defined, tested clinical flowchart — no ambiguity to model.
3. **Minimum integrations**
   No IoT sensors required for v1. Voice + basic BP manual entry + WhatsApp follow-up is sufficient.
4. **Single language first**
   Tamil-only for the pilot in Chennai-region Arogya Mandirs — cuts multilingual complexity by 80%.
5. **Measurable accuracy**
   IMCI protocol has clear red-flag criteria — easy to validate AI output against known clinical benchmarks.

THE SCENARIO — STEP BY STEP

1. **Patient arrives at Arogya Mandir**
   CHO opens Aarogya Vaani app on a basic Android phone/Webapp*.
2. **Voice triage begins**
   App speaks to patient in Native Language(English, Hindi): age, symptoms, duration, severity. NLP extracts structured data.
3. **Manual vitals entry**
   CHO enters BP & SpO2 from existing equipment. App flags danger values.
4. **SOAP note generated**
   AI produces Subjective-Objective-Assessment-Plan in under 10 seconds. CHO reviews & approves.
5. **Escalation decision**
   App recommends: Treat at home / PHC referral / Emergency.)
   Auto-drafts WhatsApp message to family (Whatsapp - Out of scope for MVP).
6. **Day-3 follow-up (Whatsapp Out of Scope for MVP)**
   Automated WhatsApp message checks patient status. Non-response flags back to CHO.

MVP SCOPE — WHAT'S IN AND WHAT'S OUT

Deliberate constraints to deliver something real in 13 weeks

✓ IN SCOPE — MVP v1

- Voice triage in Tamil (single language)
- Fever + Respiratory illness IMCI protocol
- Manual vitals entry (BP + SpO2 + Diabetic + Thermometer)
- AI-generated SOAP note (CHO review + approve)
- 3-tier escalation: Home / PHC / Emergency
- WhatsApp follow-up at Day 3 (out of scope)
- Simple web dashboard for CHO (case log) ***
- Pilot at 2 Arogya Mandirs

✗ OUT OF SCOPE — Future Phases

- Bluetooth IoT device auto-sync (Phase 2)
- Multi-language support — 5 more languages
- ABHA / National Health Stack integration
- Teleconsult slot auto-booking
- District-level analytics dashboard
- Maternal & child health use cases
- Outbreak / epidemic signal detection
- FHIR-compliant longitudinal records

TECH STACK — CHOSEN FOR SPEED, OPEN-SOURCE & CHO-GRADE RELIABILITY

All components are free/open-source or have generous free tiers — no licensing cost for MVP

FRONTEND — CHO App

- **React Native (Expo)**
  Cross-platform Android/iOS. Single codebase. Expo Go for rapid testing.
- **Voice UI via expo-speech + expo-av**
  Text-to-Speech for Tamil prompts; Audio recording for patient responses.
- **Offline-first with SQLite**
  Works in low-connectivity areas. Syncs when internet is available.

BACKEND — API & Logic

- **FastAPI (Python)**
  Lightweight REST API. Handles SOAP generation, escalation logic, and case storage.
- **PostgreSQL**
  Case records, CHO profiles, triage history. Hosted free on Supabase for MVP.
- **Celery + Redis (Whatsapp Out of scope?)**
  Async task queue for WhatsApp follow-up scheduling (day 3 triggers).

AI / NLP — Triage Engine

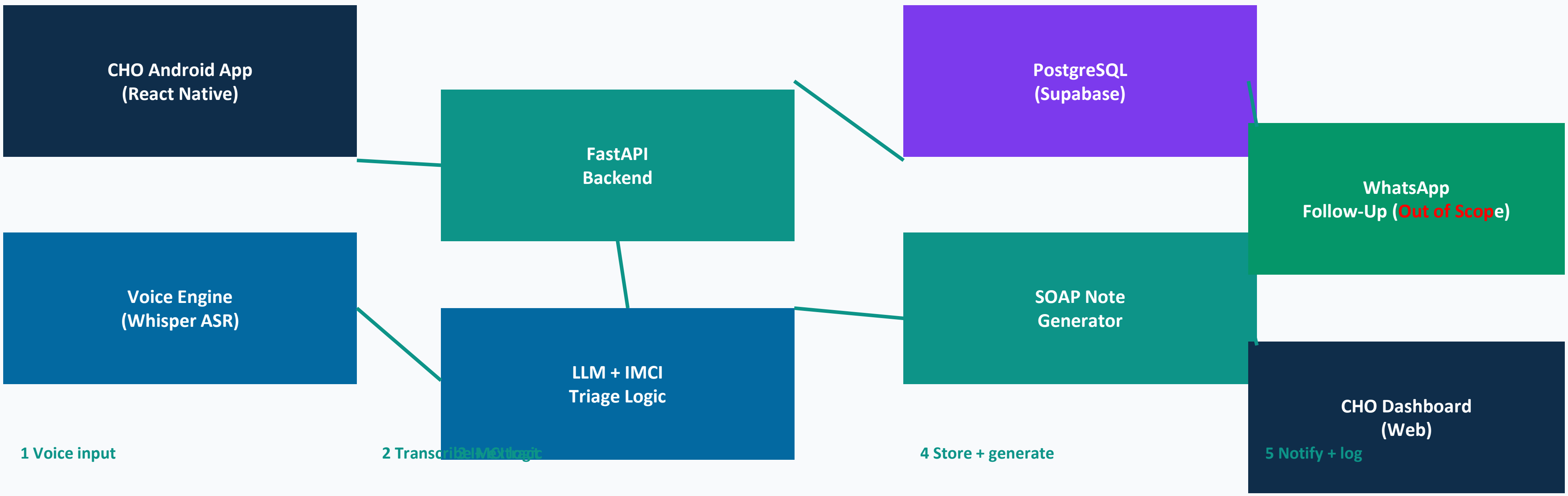
- **OpenAI Whisper (ASR)**
  State-of-the-art speech-to-text. Engish, Hindi language support. Runs on CPU.
- **GPT-4o (via API) / Llama 3 local**
  Symptom extraction + SOAP note generation. Fallback: fine-tuned Llama 3 on-device.
- **LangChain + IMCI decision tree**
  Structured clinical workflow. LLM constrained by IMCI rule-based logic for safety.

MESSAGING & INFRA

- **WhatsApp Business API (Twilio) [Whatsapp Out of scope]**
  Day-3 follow-up messages. Free 1000 conversations/month on WhatsApp Cloud API.
- **Docker + Railway / Render**
  Containerised deployment. Free tier sufficient for MVP traffic. CI/CD via GitHub Actions.
- **Supabase (Auth + Storage)**
  CHO login, case media storage. Free tier: 500MB DB, 1GB file storage.

MVP SYSTEM ARCHITECTURE

How the components connect — data flow for a single triage session



Patient → CHO app → Voice engine → FastAPI → LLM triage → SOAP note → Escalation → WhatsApp follow-up → Dashboard

13-WEEK IMPLEMENTATION PLAN | 1 Jun – 31 Aug 2026

Three sprints — each with clear team ownership and exit criteria

SPRINT 1 — Foundation

1 Jun – 28 Jun (Weeks 1–4)

Wk 1–2	Clinical protocol mapping — digitise IMCI fever/ARI decision tree (M5+M6)
Wk 1–2	App scaffolding — React Native Expo project, navigation, auth screens (M3+M4)
Wk 2–3	Whisper ASR integration — Tamil STT working on test phrases (M3+M4)
Wk 3–4	FastAPI backend + PostgreSQL schema on Supabase (M7+M1)
Wk 4	Voice prompt script writing in Tamil with clinical advisor (M5+M6)

EXIT ✓

Voice app records patient speech → transcribes → returns structured symptom JSON

SPRINT 2 — Build Core

29 Jun – 26 Jul (Weeks 5–8)

Wk 5–6	IMCI triage engine — LangChain + GPT-4o with IMCI rules as constraints (M3+M4)
Wk 5–6	SOAP note generator — prompt engineering + CHO approval flow (M3+M4)
Wk 6–7	Manual vitals entry UI + danger-value alert logic (M5+M6)
Wk 7–8	Escalation output (3-tier) + WhatsApp API integration via Twilio (M7+M1)
Wk 8	End-to-end integration test — full triage session on staging (All pairs)

EXIT ✓

Full triage session completes in < 5 min. SOAP note generated. Escalation sent.

SPRINT 3 — Pilot & Validate

27 Jul – 31 Aug (Weeks 9–13)

Wk 9	CHO onboarding training at 2 Arogya Mandirs (M1+M2, M5+M6)
Wk 9–10	Live pilot — 20+ real triage sessions per site (All pairs)
Wk 10–11	Accuracy validation — AI vs clinical advisor review of SOAP notes (M5+M6)
Wk 11–12	Bug fixes, CHO feedback incorporation, WhatsApp follow-up check (M3+M4)
Wk 13	Final report, demo video, capstone presentation (M1+M2)

EXIT ✓

≥85% SOAP accuracy. CHO satisfaction ≥4/5. Pilot data presented.

TEAM OWNERSHIP — WHO BUILDS WHAT

Mapped to the 4 paired workstreams from the project plan

M1 + M2
Project Lead & Strategy

- Sprint planning, weekly standups, risk log
- CHO onboarding coordination at pilot sites
- Final capstone report & demo presentation
- Stakeholder communication & milestone sign-off

PM across all sprints · Sprint 3: pilot logistics & delivery

M3 + M4
AI/ML, NLP & Voice UX

- Whisper ASR integration — Tamil speech-to-text
- LangChain + GPT-4o IMCI triage engine
- SOAP note generator — prompt engineering
- React Native voice UI & triage conversation flow

Sprint 1: ASR + app scaffold · Sprint 2: triage engine + SOAP · Sprint 3: accuracy testing

M5 + M6
IoT Integration & Clinical

- IMCI fever/ARI protocol — digitise decision tree
- Manual vitals entry UI + danger-value alerts
- Clinical validation of AI SOAP note output
- Hindi/English voice prompt script with clinical accuracy

Sprint 1: protocol mapping · Sprint 2: vitals + alerts · Sprint 3: clinical validation

M7 + M1
Data Architecture & Follow-Up

- FastAPI backend + PostgreSQL schema (Supabase)
- Case record storage & retrieval API
- WhatsApp follow-up workflow — Twilio integration (Out of scope)
- CHO web dashboard — case log & status view

Sprint 1: DB schema + API · Sprint 2: WhatsApp + dashboard · Sprint 3: data reporting

SUCCESS METRICS & KEY RISKS

How we measure MVP success and what could go wrong

MVP SUCCESS METRICS

≥85%
SOAP note clinical accuracy vs clinical advisor review

<5 min
End-to-end triage session duration per patient

≥4/5
CHO satisfaction score post-pilot survey

40+
Real triage sessions completed at pilot sites

Day 3
Follow-up message sent & non-responders flagged

RISKS & MITIGATIONS

! **Whisper ASR accuracy in Tamil dialects**
→ Pre-test on 5 CHO voices; fine-tune on domain words; add manual correction option

! **GPT-4o API costs for 40+ sessions**
→ Cache common IMCI patterns; use Llama 3 local as fallback; budget ₹5k cap

! **CHO reluctance to use a new tool**
→ Co-design 2 sessions before pilot; keep UI under 3 taps to any action

! **WhatsApp approval for business API**
→ Apply Week 1; use Twilio sandbox for testing; SMS as backup

! **Network outage at Arogya Mandir**
→ SQLite offline cache; auto-sync when connected; no data lost

THE PLAN IN THREE SENTENCES

Build a voice-first Hindi/English triage app for fever and respiratory illness.

Test it live with real CHOs at 2 Arogya Mandirs.

Prove ≥85% clinical accuracy. Then scale.

1 Jun	**13 Wks**	**HINDI/ENGLISH**	**31 Aug**
Start	Duration	Language v1	Delivery